

Bootstrap — Topics 15-17: Images, Tables & Forms

Study paper format: notes (Roman Urdu + English technical terms) + active recall Q&A at the end. Covers: Images, Tables, Forms (★ ★ ★).

15. Images

Bootstrap images ko responsive aur styled banane ke liye chand simple utility classes deta hai.

Responsive Image

```
img-fluid
```

Image ko apne parent container ke andar **responsive** bana deta hai — `max-width: 100%` aur `height: auto` set karta hai, isliye image kabhi apne container se overflow nahi hoti aur different screen sizes par properly scale hoti hai.

```
html


```

Thumbnail

```
img-thumbnail
```

Image ke around ek halka border, padding, aur rounded corners add karta hai — jaise ek photo frame.

```
html


```

Rounded

```
rounded-circle
```

Image ko perfect **circle** shape de deta hai (profile pictures/avatars ke liye common) — image ki width aur height barabar honi chahiye taake circle theek ban sake.

```
html
```

```

```

Combine bhi kar sakte ho:

```
html
```

```

```

16. Tables

Bootstrap `.table` class se plain HTML tables ko quickly stylish bana deta hai.

Basic Table

```
html
```

```
<table class="table">
  <thead>
    <tr>
      <th>Name</th>
      <th>Email</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Ali</td>
      <td>ali@example.com</td>
    </tr>
  </tbody>
</table>
```

`.table` class basic spacing, borders (bottom lines), aur clean typography deti hai.

Striped

```
html
<table class="table table-striped">
```

Har alternate row ko light background color deta hai — reading easier hoti hai lambay tables mein.

Hover

```
html
<table class="table table-hover">
```

Row par mouse hover karne par uska background highlight ho jata hai.

Dark

```
html

<table class="table table-dark">
```

Poore table ko dark theme (dark background, light text) deta hai. Ye striped/hover ke sath bhi combine ho sakta hai: `table table-dark table-striped`.

Responsive Tables

```
html

<div class="table-responsive">
  <table class="table">
    ...
  </table>
</div>
```

Agar table chhoti screens par fit na ho, to `.table-responsive` wrapper table ko **horizontally scrollable** bana deta hai — content cut nahi hota, bas user side-scroll kar sakta hai.

Note: `.table-responsive` khud table par nahi, table ke **wrapping** `<div>` par lagti hai.

17. Forms ★ ★ ★

Bootstrap forms ke elements — inputs, selects, checkboxes, etc. — ko consistent aur professional look deta hai.

Inputs

```
html
```

```
<div class="mb-3">
  <label for="name" class="form-label">Name</label>
  <input type="text" class="form-control" id="name" placeholder="Enter your
</div>
```

`.form-control` class text inputs ko full-width, properly padded, aur styled bana deti hai.

`.form-label` label ko consistent spacing deti hai.

Textarea

```
html

<div class="mb-3">
  <label for="message" class="form-label">Message</label>
  <textarea class="form-control" id="message" rows="3"></textarea>
</div>
```

Textarea par bhi `.form-control` hi use hoti hai — same styling text input jaisi.

Select

```
html

<select class="form-select">
  <option selected>Choose an option</option>
  <option value="1">One</option>
  <option value="2">Two</option>
</select>
```

Dropdown ke liye `.form-select` class use hoti hai (`.form-control` nahi).

Checkbox

```
html
```

```
<div class="form-check">
  <input class="form-check-input" type="checkbox" id="terms">
  <label class="form-check-label" for="terms">
    I agree to the terms
  </label>
</div>
```

`.form-check` wrapper div hai, `.form-check-input` checkbox ko style karti hai, `.form-check-label` label ko align karti hai.

Radio

```
html

<div class="form-check">
  <input class="form-check-input" type="radio" name="plan" id="basic">
  <label class="form-check-label" for="basic">Basic</label>
</div>
<div class="form-check">
  <input class="form-check-input" type="radio" name="plan" id="pro">
  <label class="form-check-label" for="pro">Pro</label>
</div>
```

Same classes checkbox jaisi, bas `type="radio"` aur same `name` attribute (mutually exclusive selection ke liye).

Switch

```
html
```

```
<div class="form-check form-switch">
  <input class="form-check-input" type="checkbox" id="notifications">
  <label class="form-check-label" for="notifications">Enable Notifications</label>
</div>
```

`.form-check` ke sath extra `.form-switch` class checkbox ko ek **toggle switch** jaisa look deti hai (On/Off style).

Floating Labels

```
html

<div class="form-floating mb-3">
  <input type="email" class="form-control" id="emailInput" placeholder="name@example.com">
  <label for="emailInput">Email address</label>
</div>
```

`.form-floating` wrapper input ke andar label ko **"float"** up karva deta hai jab user click/type kare — modern, compact look, especially clean forms ke liye popular.

Input Groups

```
html

<div class="input-group mb-3">
  <span class="input-group-text">@</span>
  <input type="text" class="form-control" placeholder="Username">
</div>
```

`.input-group` input ke sath ek extra element (icon, text, button) attach karne deta hai — jaise `@` symbol, currency sign, ya search button.

```
html
```

```
<div class="input-group">
  <input type="text" class="form-control" placeholder="Search">
  <button class="btn btn-primary" type="button">Search</button>
</div>
```

Validation

html

```
<input type="text" class="form-control is-valid">
<div class="valid-feedback">Looks good!</div>

<input type="text" class="form-control is-invalid">
<div class="invalid-feedback">Please provide a valid value.</div>
```

`.is-valid` / `.is-invalid` classes input ko green/red border aur icon dete hain. `.valid-feedback` / `.invalid-feedback` corresponding message dikhate hain (jab input already validated ho tab automatically show hote hain).

Range Slider

html

```
<label for="volume" class="form-label">Volume</label>
<input type="range" class="form-range" id="volume" min="0" max="100">
```

`.form-range` class range/slider inputs ko style karti hai.

Disabled Inputs

html

```
<input type="text" class="form-control" placeholder="Disabled input" disabled>
```



disabled HTML attribute input ko grayed-out aur non-interactive bana deta hai — value form submit ke sath bhi nahi jati.

Readonly

html

```
<input type="text" class="form-control" value="Cannot be edited" readonly>
```

readonly attribute input ko edit hone se rokta hai, lekin normal look mein rehta hai (disabled jaisa gray nahi), aur value form submit ke sath jati hai.

Disabled vs Readonly:

	Disabled	Readonly
Look	Grayed-out	Normal
Editable	Nahi	Nahi
Form submit mein value	Nahi jati	Jati hai
Focusable	Nahi	Haan

Cheat Sheet

Section	Key Classes
Images	<code>img-fluid</code> , <code>img-thumbnail</code> , <code>rounded-circle</code>
Tables	<code>table</code> , <code>table-striped</code> , <code>table-hover</code> , <code>table-dark</code> , <code>table-responsive</code>
Text Inputs/Textarea	<code>form-control</code> , <code>form-label</code>
Select	<code>form-select</code>
Checkbox/Radio	<code>form-check</code> , <code>form-check-input</code> , <code>form-check-label</code>
Switch	<code>form-check</code> <code>form-switch</code>
Floating Labels	<code>form-floating</code>
Input Group	<code>input-group</code> , <code>input-group-text</code>
Validation	<code>is-valid</code> , <code>is-invalid</code> , <code>valid-feedback</code> , <code>invalid-feedback</code>
Range	<code>form-range</code>
Disabled/Readonly	<code>disabled</code> , <code>readonly</code> (HTML attributes, not Bootstrap classes)

Practice Exercises

1. Ek profile card banao jisme `img-fluid rounded-circle` wali avatar image ho, aur ek `table-striped table-hover` wali skills table ho.
2. Ek complete signup form banao: text input (name), email input with floating label, password input, aur ek terms checkbox — sab `form-control`/`form-check` classes ke

sath.

3. Ek search bar banao `input-group` use kar ke jisme input ke sath ek search icon/button attached ho.
 4. Ek form field banao jo `is-invalid` state show kare aur uske sath `invalid-feedback` message ho.
 5. Table ko `table-responsive` wrapper mein wrap karo aur observe karo ke chhoti screen par ye kaise scroll karti hai.
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Active Recall — Images, Tables & Forms

┆ Khud se answer karne ki koshish karo, phir neeche scroll karke check karo.

- Q1. `img-fluid` class kya karti hai, aur ye kaunse do CSS properties set karti hai?
 - Q2. `img-thumbnail` aur `rounded-circle` mein kya farq hai?
 - Q3. `table-responsive` class kaha lagai jati hai — table par ya kisi wrapper par?
 - Q4. `table-striped` aur `table-hover` mein kya farq hai?
 - Q5. Text input aur select dropdown mein alag-alag konsi classes use hoti hain?
 - Q6. Checkbox ko toggle switch jaisa dikhane ke liye konsi extra class add karni parti hai?
 - Q7. `.form-floating` kya effect deta hai?
 - Q8. `input-group` kis liye use hota hai? Ek example do.
 - Q9. Form validation ke liye konsi classes use hoti hain, aur `invalid-feedback` kab show hota hai?
 - Q10. `disabled` aur `readonly` attributes mein main differences kya hain?
-

Answers

<details> <summary>Answers dekhne ke liye click karein</summary>

A1. `img-fluid` image ko responsive banati hai — `max-width: 100%` aur `height: auto` set karti hai, isliye image parent container se overflow nahi hoti aur proportionally scale hoti hai.

A2. `img-thumbnail` image ke around border, padding, aur halke rounded corners add karta hai (frame jaisa look). `rounded-circle` image ko perfect circle shape deta hai (avatars ke liye).

A3. `table-responsive` table ke **wrapping** `<div>` par lagai jati hai, khud table element par nahi — jaise `<div class="table-responsive"><table class="table">...</table></div>`.

A4. `table-striped` alternate rows ko light background color deta hai (readability ke liye). `table-hover` row par mouse hover karne par uska background highlight karta hai (interactivity ke liye).

A5. Text input aur textarea ke liye `.form-control` use hoti hai, jabke select dropdown ke liye `.form-select` use hoti hai.

A6. `.form-check` ke sath extra `.form-switch` class add karni parti hai — `<div class="form-check form-switch">`.

A7. `.form-floating` input ke andar placeholder/label ko user ke type/click karne par upar "float" karva deta hai — ek compact, modern label style hai.

A8. `input-group` input field ke sath extra element (icon, text prefix/suffix, ya button) attach karne ke liye use hota hai. Example: `@` symbol username input ke sath, ya search button search input ke sath.

A9. `.is-valid` / `.is-invalid` classes input par lagai jati hain (green/red border + icon), aur `.valid-feedback` / `.invalid-feedback` corresponding message dikhati hain. `invalid-feedback` tab show hota hai jab input `is-invalid` class ke sath ho (ya form validation ke baad).

A10. `disabled` input ko grayed-out aur completely non-interactive banata hai, aur uski value form submit ke sath nahi jati. `readonly` input ko normal look mein rakhta hai (gray

nahi), user edit nahi kar sakta, lekin value form submit ke sath jati hai aur input focusable rehta hai.

</details>